Operating Systems

Final Course Project

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Introduction

The focus of this project was to familiarize myself with the key features of the Linux operating system by creating a virtual machine for the IoT.

This venture demonstrates my knowledge of the Linux file system hierarchy, ability to create shell scripts, ability to run administrative tasks, ability to create basic network configurations, and ability to identify/mitigate security risks.

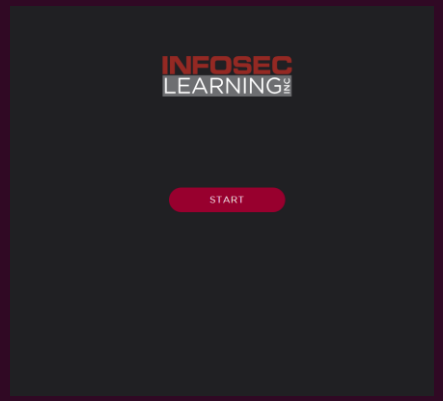


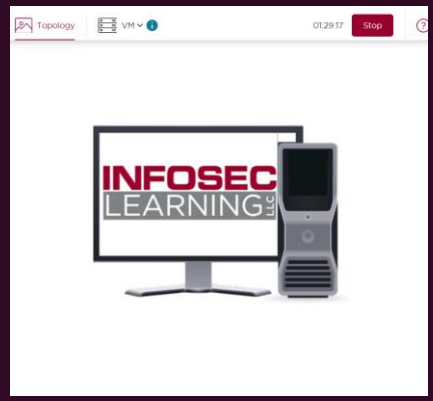
Linux Operating System

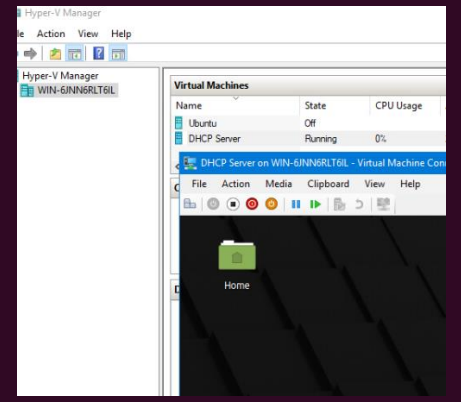
Linux is a portable open-source lightweight multi-tasking multi-user UNIX operating system. It has a graphical user interface yet can operate solely on the command line interface if desired. It provides a hierarchical file system, authentication features for user security, and an interpreter program called the Shell which executes commands of the operating system.

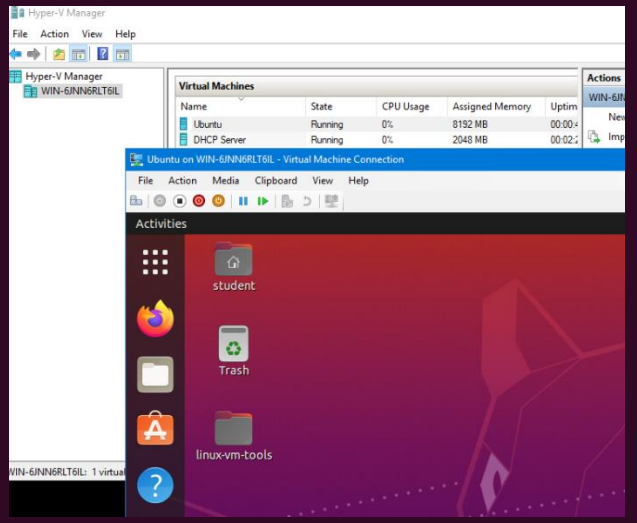
Launching and Shutting Down a Virtual Machine

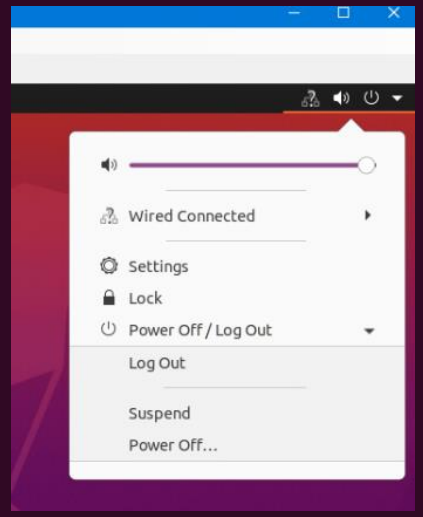
The provided imagery demonstrates my ability to launch, access and shut down a virtual machine that uses the Ubuntu Linux operating system via a DHCP server.



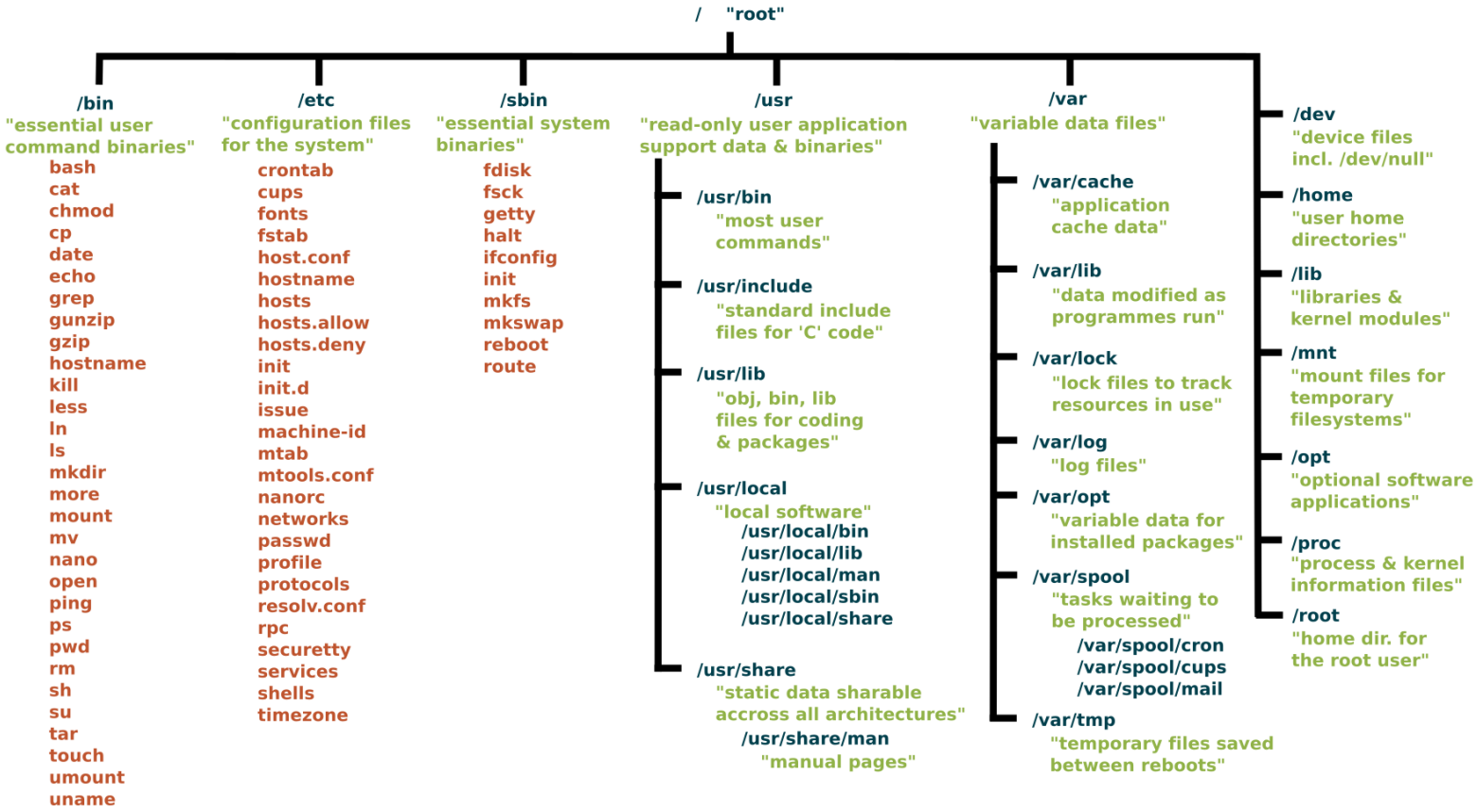








Linux Filesystem Overview



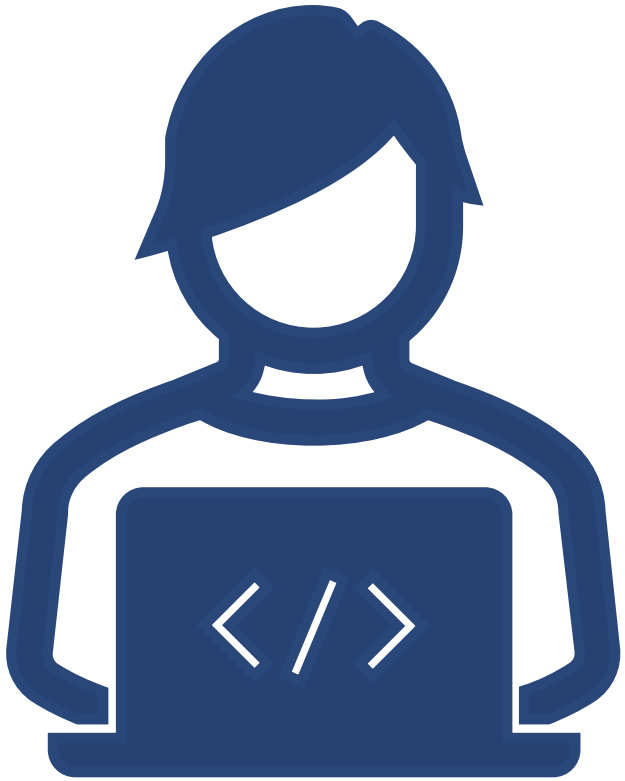


Linux Filesystem

The Linux directory structure uses both relative and absolute pathnames.

A relative path is the location of a file relative to its working directory and an absolute path is the exact location of a file that always starts from the root directory (/).

The Linux file system allows the modification of a file and directory ownership by identifying, defining, and modifying its permissions.



Shell Scripts

Executing BASH shell scripts allows an individual to perform a series of actions such as creating a folder, navigating to a specific directory, viewing or terminating a process, and schedule commands to be executed in the future.

Some basic shell script commands are:

- The PWD (Print Working Directory) command, which displays the full path of the current working directory.
- The CD (Change Directory) command, which allows an individual to switch from one working directory to another.

The Linux shell can also manipulate command input and output by using redirection and pipe meta-characters.

Creation of Directories and Files

```
student@ubuntu1:~$ cd ~
student@ubuntu1:~$ pwd
/home/student
student@ubuntu1:~$ mkdir JanFebSession
student@ubuntu1:~$ cd JanFebSession
student@ubuntu1:~/JanFebSession$ mkdir Course1
student@ubuntu1:~/JanFebSession$ mkdir Course2
student@ubuntu1:~/JanFebSession$ mkdir Course3
student@ubuntu1:~/JanFebSession$ cd Course1
student@ubuntu1:~/JanFebSession/Course1$ touch file1 file2 file3
student@ubuntu1:~/JanFebSession/Course1$ tree -d -L 2 ~
/home/student
├── Desktop
│   └── linux-vm-tools
├── Documents
├── Downloads
├── JanFebSession
│   ├── Course1
│   ├── Course2
│   └── Course3
├── Music
├── Pictures
├── Public
├── shared-drives
├── snap
│   └── snap-store
├── Templates
└── Videos

16 directories
student@ubuntu1:~/JanFebSession/Course1$ ls -l ~/JanFebSession/Course1
total 0
-rw-rw-r-- 1 student student 0 Nov  2 01:03 file1
-rw-rw-r-- 1 student student 0 Nov  2 01:03 file2
-rw-rw-r-- 1 student student 0 Nov  2 01:03 file3
student@ubuntu1:~/JanFebSession/Course1$ echo "Francisco Acosta"
Francisco Acosta
student@ubuntu1:~/JanFebSession/Course1$ date
Thu 02 Nov 2023 01:05:54 AM EDT
student@ubuntu1:~/JanFebSession/Course1$
```

Copying and Removing Directories and Files

```
student@ubuntu1:~$ rmdir MarAprSession/Course3
student@ubuntu1:~$ rm MarAprSession/Course1/file3
student@ubuntu1:~$ tree JanFebSession MarAprSession
JanFebSession
├── Course1
│   ├── file1
│   ├── file2
│   └── file3
├── Course2
└── Course3
MarAprSession
├── Course1
│   ├── file1
│   └── file2
└── Course2

5 directories, 5 files
student@ubuntu1:~$ echo "Francisco Acosta"
Francisco Acosta
student@ubuntu1:~$ date
Thu 02 Nov 2023 01:15:46 AM EDT
student@ubuntu1:~$
```

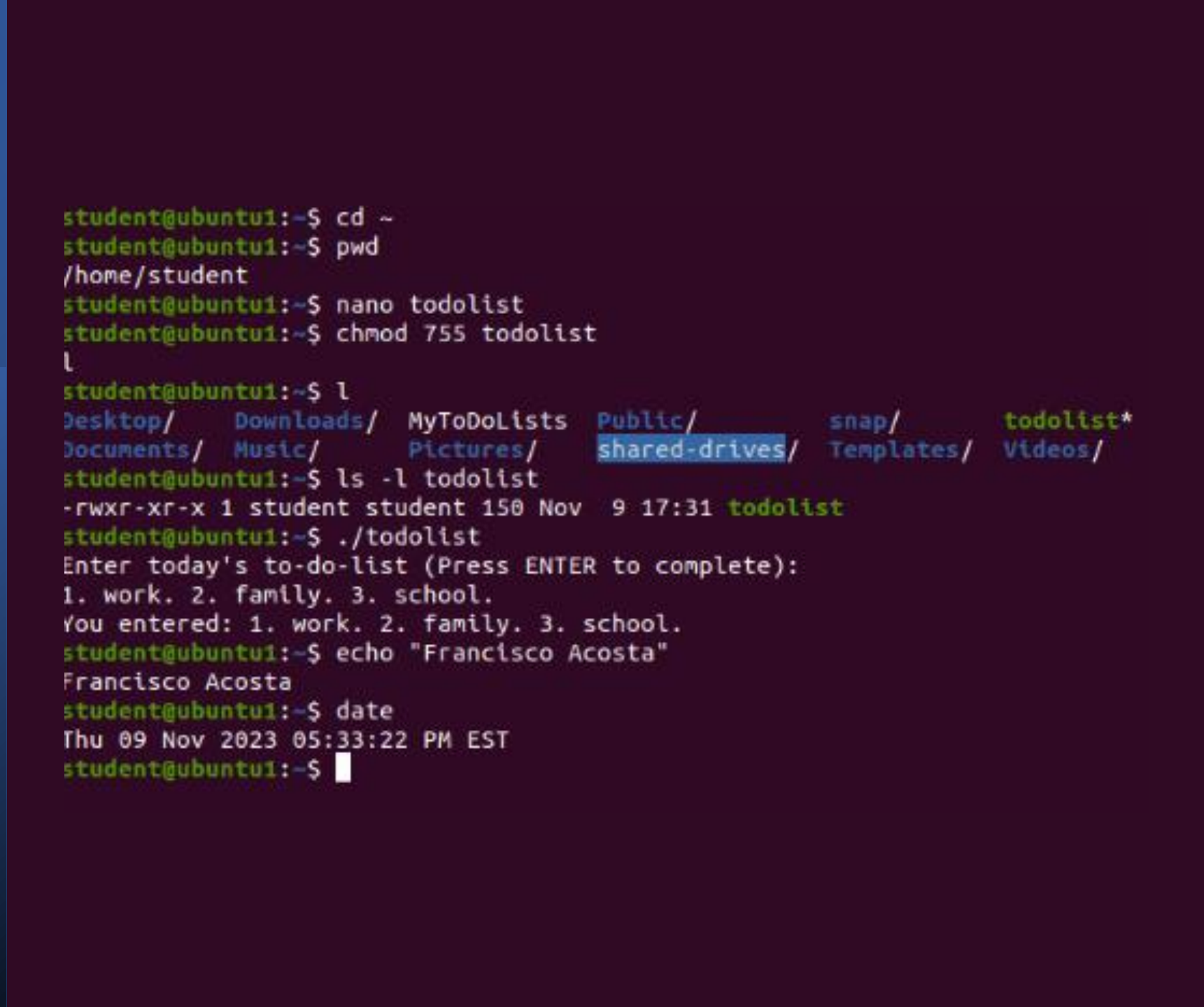
Locating Directories and Files

```
student@ubuntu1:~$ cd ~
student@ubuntu1:~$ pwd
/home/student
student@ubuntu1:~$ find ~ -iname course* -type d
/home/student/JanFebSession/Course2
/home/student/JanFebSession/Course1
/home/student/JanFebSession/Course3
student@ubuntu1:~$ find ~ -iname file1
/home/student/JanFebSession/Course1/file1
student@ubuntu1:~$ find ~ -iname file*
/home/student/JanFebSession/Course1/file3
/home/student/JanFebSession/Course1/file1
/home/student/JanFebSession/Course1/file2
student@ubuntu1:~$ locate -S
Database /var/lib/mlocate/mlocate.db:
        39,922 directories
        448,889 files
        31,465,051 bytes in file names
        11,284,336 bytes used to store database
student@ubuntu1:~$ sudo updatedb
[sudo] password for student:
student@ubuntu1:~$ locate -i course
/home/student/JanFebSession/Course1
/home/student/JanFebSession/Course2
/home/student/JanFebSession/Course3
/home/student/JanFebSession/Course1/file1
/home/student/JanFebSession/Course1/file2
/home/student/JanFebSession/Course1/file3
student@ubuntu1:~$ locate -r /file1$
/home/student/JanFebSession/Course1/file1
student@ubuntu1:~$ echo "Francisco Acosta"
Francisco Acosta
student@ubuntu1:~$ date
Thu 02 Nov 2023 12:41:26 AM EDT
student@ubuntu1:~$
```

Shell Scripts

- **The permissions for a file are read, write and execute.**
  They may be applied to a user, a group or everyone else.
  The below example gives the user/group read and write permissions.
  Everyone else only has read permissions.
  Example: -rw-rw-r--
- **The name of a user-defined variable in a script is an alpha-numerical value, such as "Text".**
- **A meta-character may be used in a script to redirect the input/output to or from a file.**
  An example is the >> meta-character, which redirects to a file and appends to the end of it.

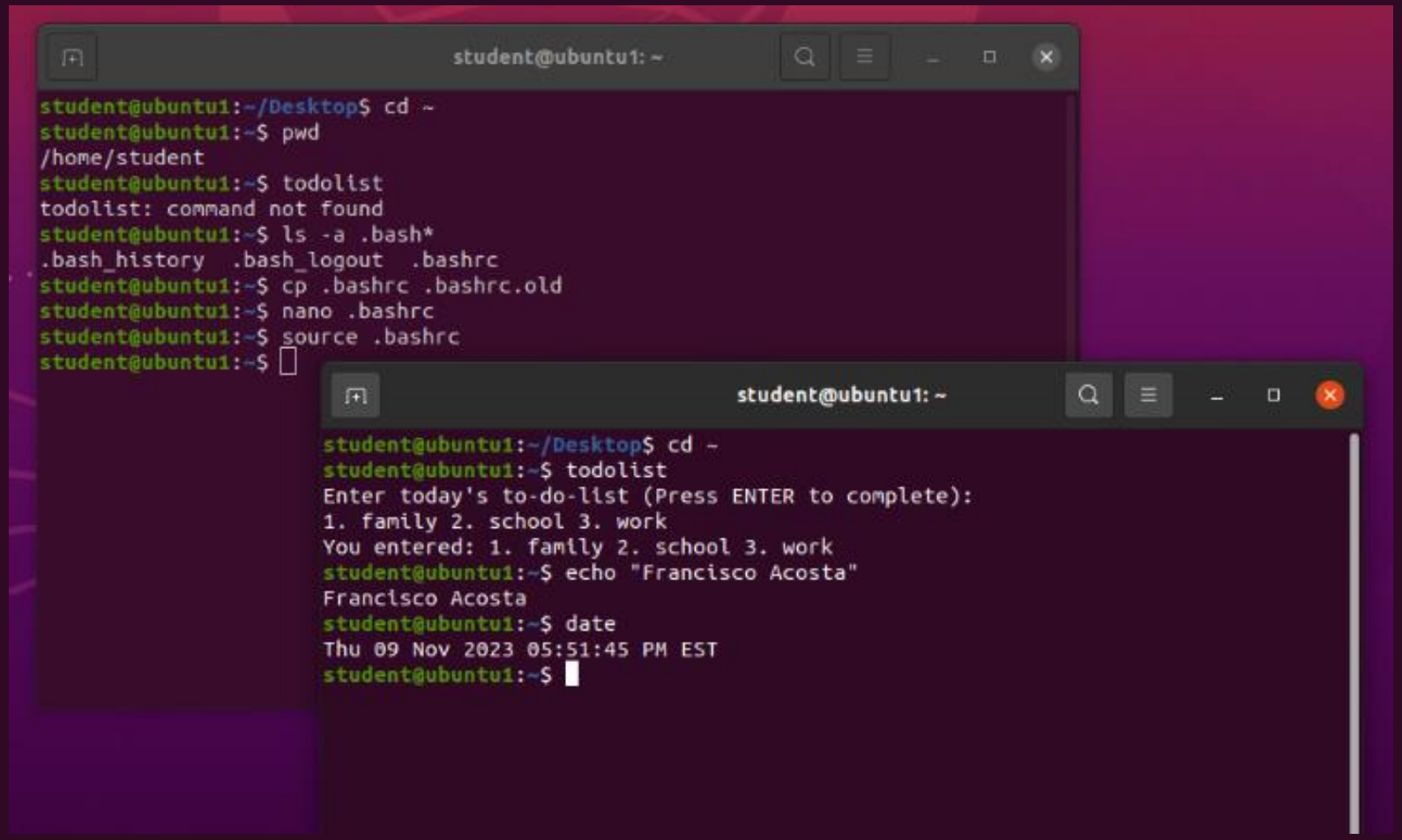
Changing File Permissions Script

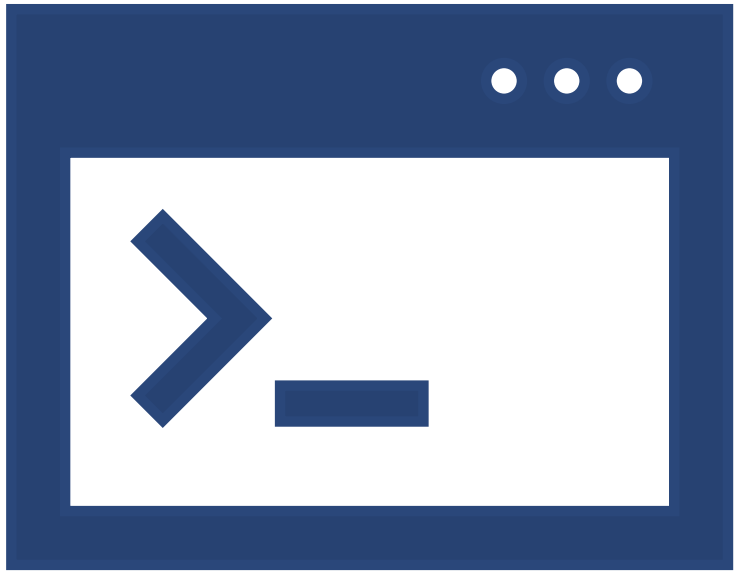


Setting the PATH Variable

```
student@ubuntu1:~$ cd ~
student@ubuntu1:~$ pwd
/home/student
student@ubuntu1:~$ todolist
todolist: command not found
student@ubuntu1:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:
al/games:/snap/bin
student@ubuntu1:~$ PATH=$PATH:/home/student
student@ubuntu1:~$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:
al/games:/snap/bin:/home/student
student@ubuntu1:~$ todolist
Enter today's to-do-list (Press ENTER to complete):
1. school. 2. work. 3. family.
You entered: 1. school. 2. work. 3. family.
student@ubuntu1:~$ echo "Francisco Acosta"
Francisco Acosta
student@ubuntu1:~$ date
Thu 09 Nov 2023 05:37:42 PM EST
student@ubuntu1:~$
```

Making the PATH Variable Permanent





Linux Administration Tasks

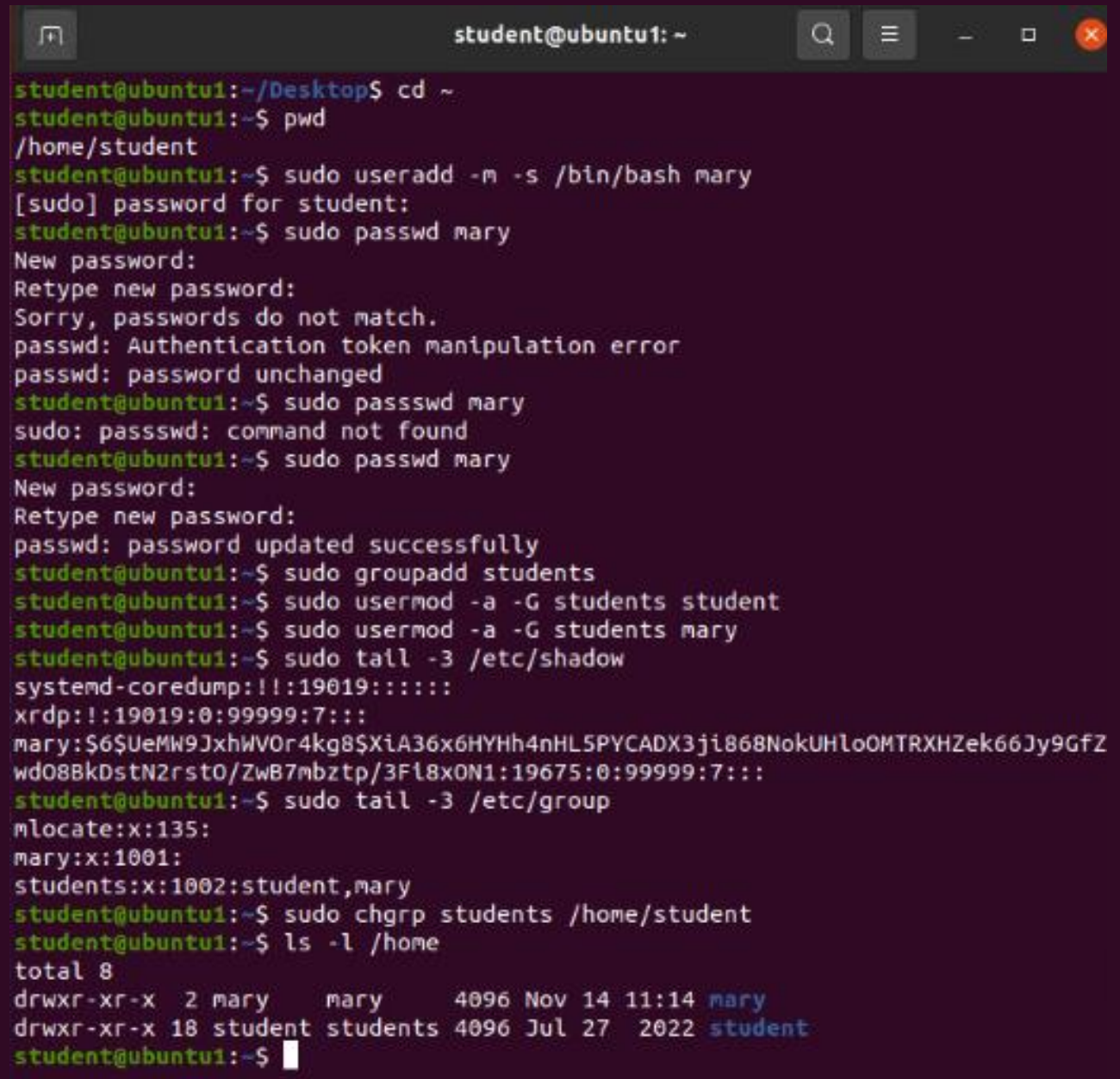
Linux administration tasks include user and group management such as creating, modifying, and deleting accounts. They also include managing log files, automation, installing software, identifying security vulnerabilities, and many more.

Adding Users and Groups

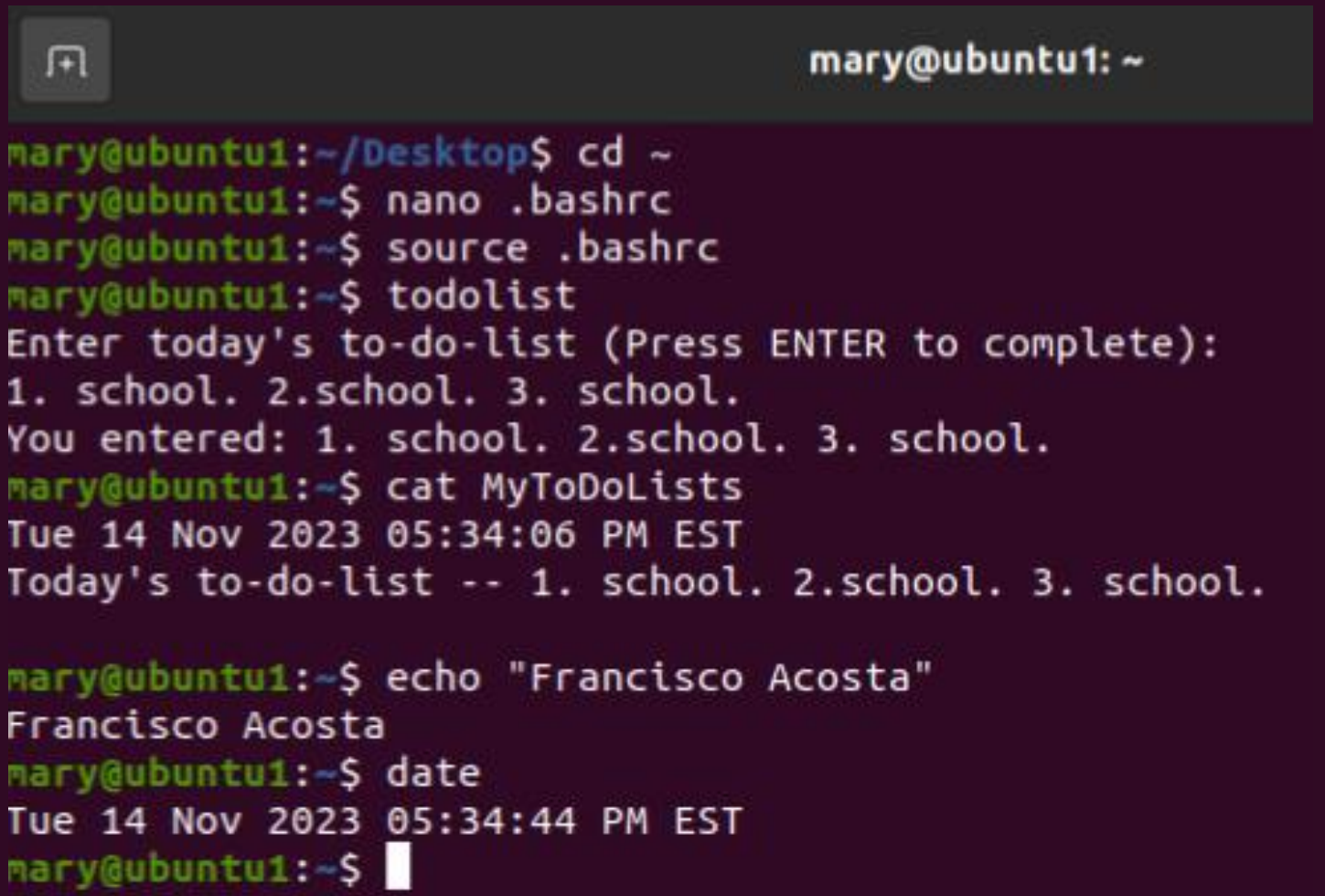
The addition of users and groups may be added using the Linux Graphical User Interface (GUI).

Additions may also be managed by Linux's Command-Line Interface (CLI).
The CLI has many options when creating such accounts. An example is the "–m" option in the useradd command which creates a home directory for a new user. Another example is the "-3" option in the tail command that tells how many lines to show from the end of the file.

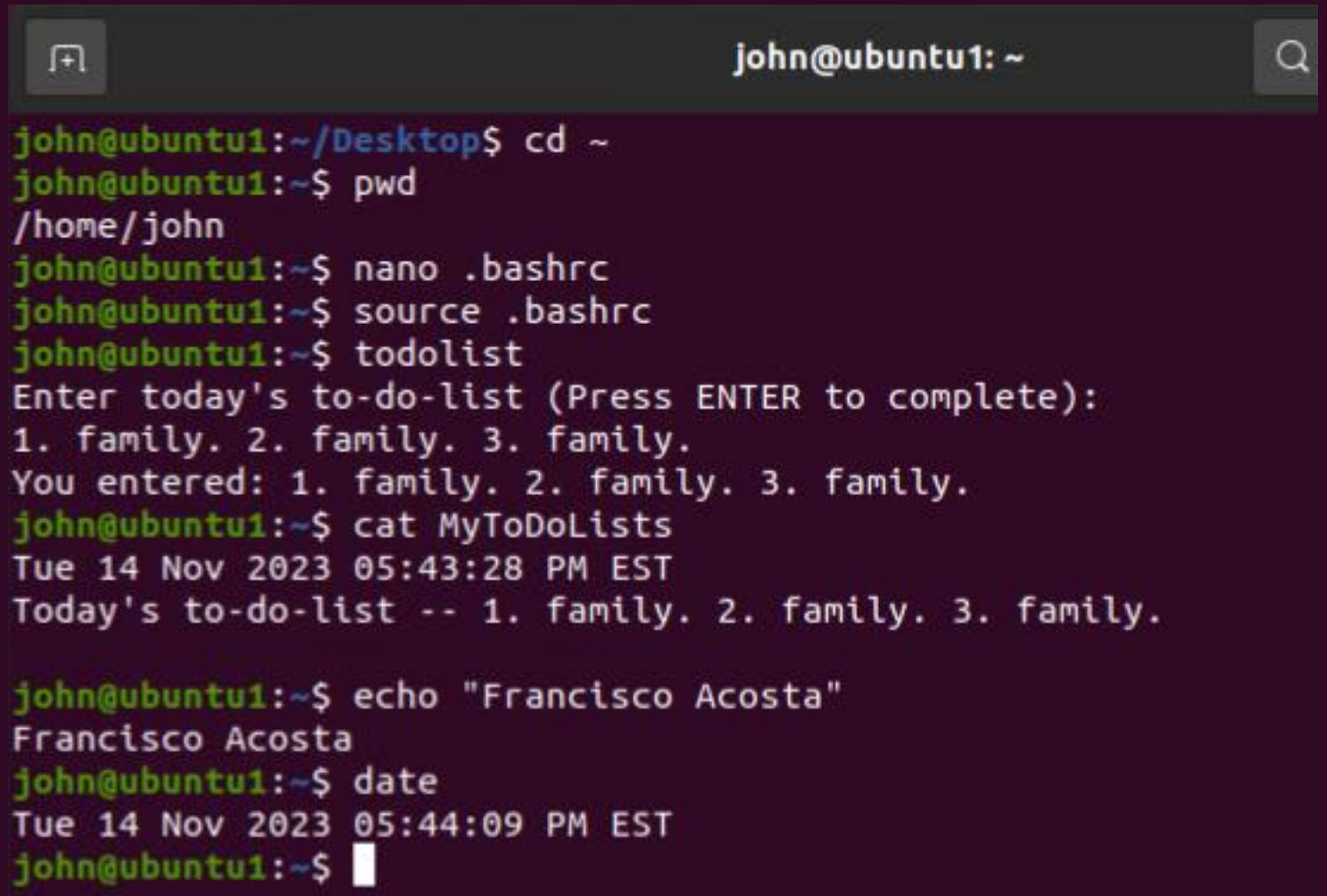
Adding Users with the CLI



Testing User and Group Settings

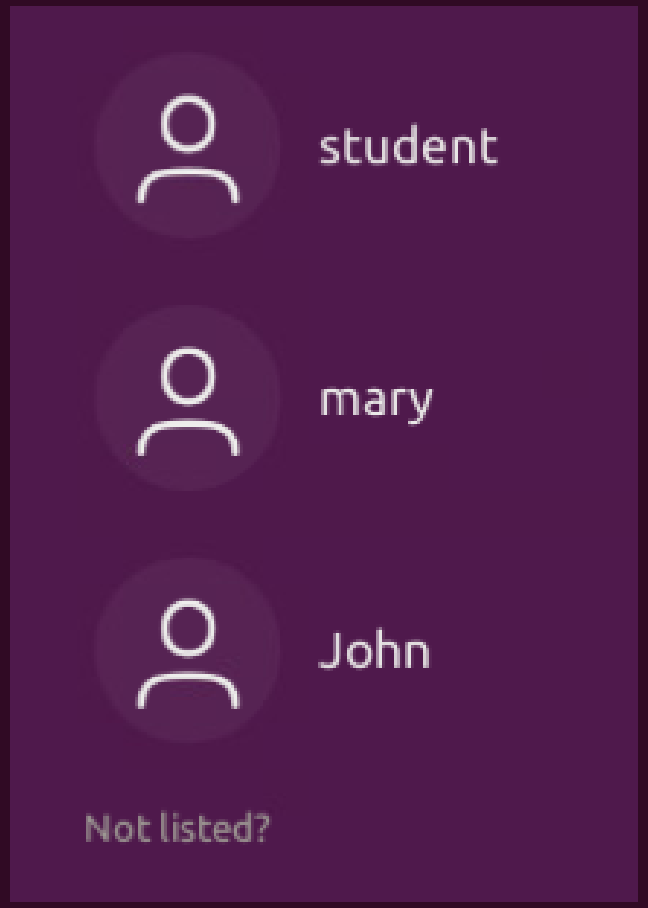


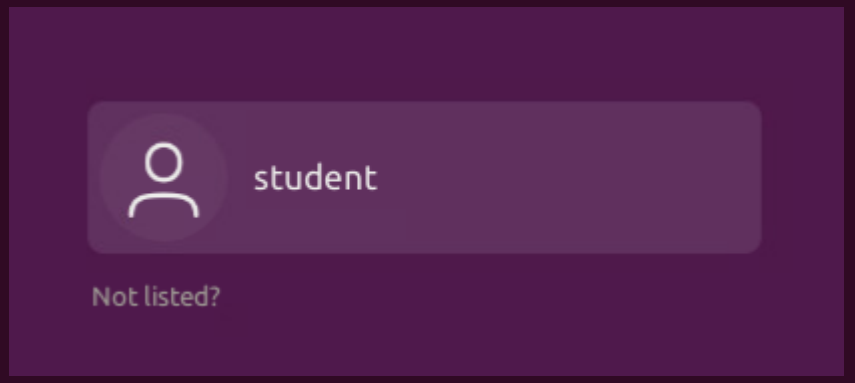
Adding Users with the GUI

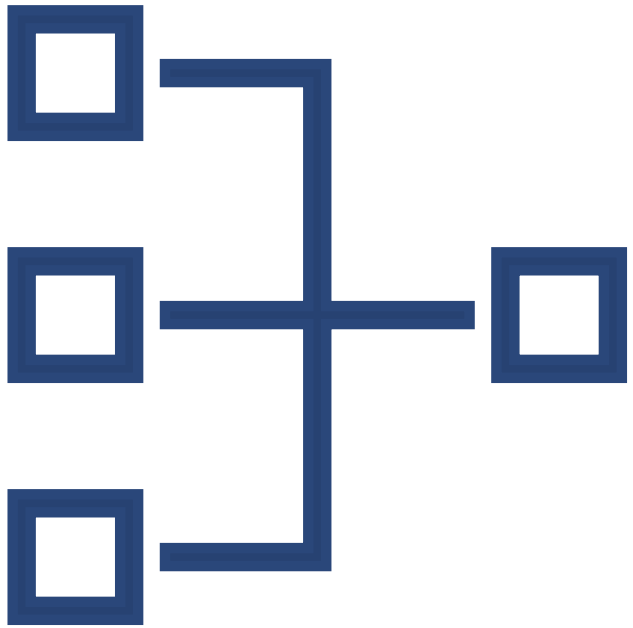


Removing Users and Groups

The provided imagery demonstrates the removal of two users within the Linux operating system. The same principles may be applied to a group.







Network Configuration

This project gave me the knowledge to configure the Linux operating system so it may participate on a network. Configurations include network interface, domain name space, and IP routing.

I am also able to examine the configuration of network services that provide infrastructure, remote administration, web, file sharing, e-mail, and database services to users across a network.

I was also fortunate to learn how network services and apps are hosted within cloud environments.

Discovering Host IP Configurations

I was able to find:

- **the IP address of my Ubuntu machine**
  - ✓ **192.168.1.107**
- **the IP address of the default gateway**
  - ✓ **192.168.1.1**
- **the IP address of the DHCP server**
  - ✓ **192.168.1.1**
- **the IP address of the DNS server**
  - ✓ **192.168.1.1**

```
student@ubuntu1:/var/lib/dhcp$ ping -c 4 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.413 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.516 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.330 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=0.572 ms

--- 192.168.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3067ms
rtt min/avg/max/mdev = 0.330/0.457/0.572/0.093 ms
student@ubuntu1:/var/lib/dhcp$ echo "Francisco Acosta"
Francisco Acosta
student@ubuntu1:/var/lib/dhcp$ date
Mon 04 Sep 2023 10:07:20 AM EDT
student@ubuntu1:/var/lib/dhcp$
```

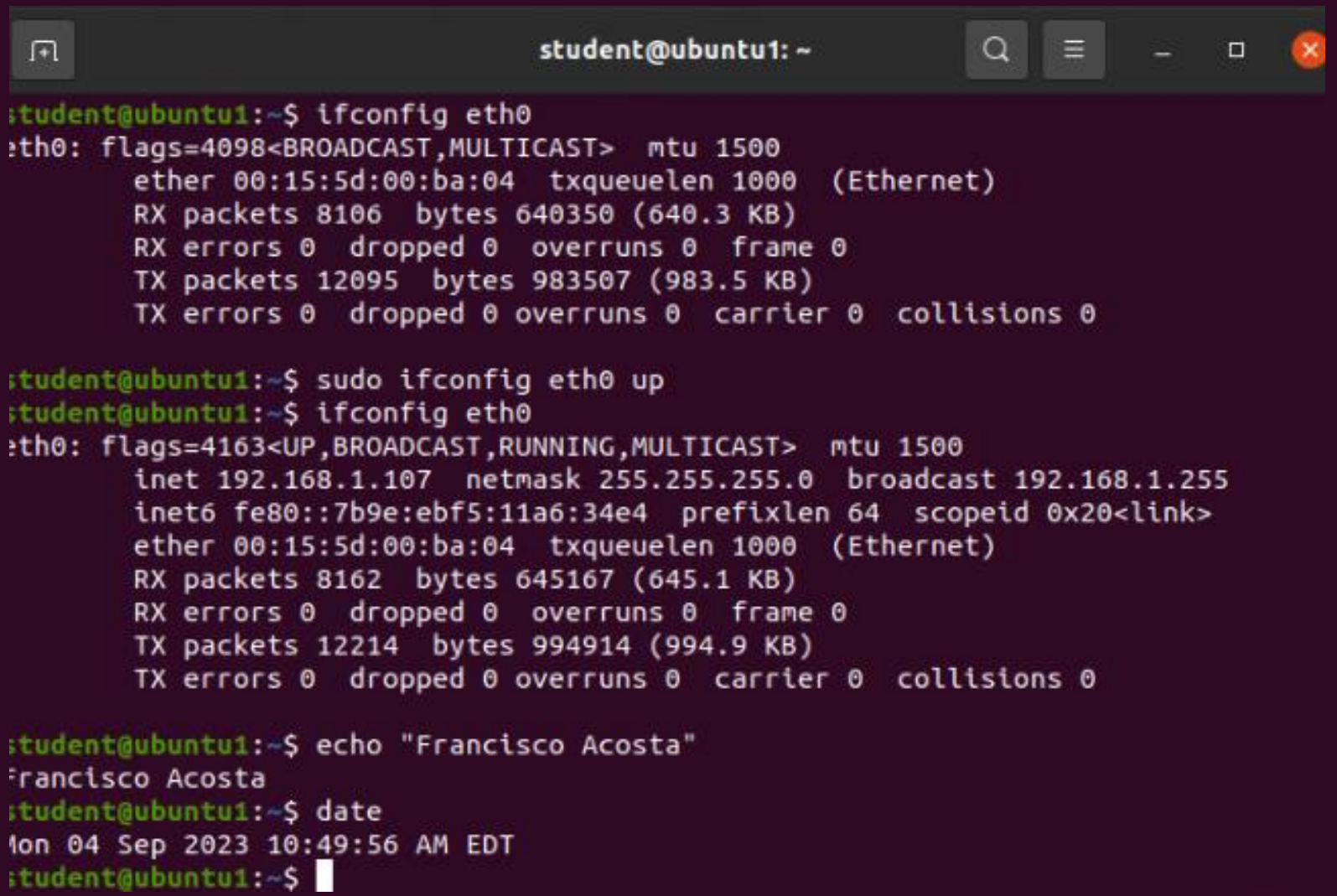
Managing Network Interfaces

During the management of network interfaces, I found the DHCP message shown in the output of the "sudo dhclient –v –r ens33" command was "DHCPRELEASE".

I also found four DHCP messages shown in the output of the "sudo dhclient –v ens33" command. The four messages are as follows:

- **DHCPDISCOVER**
- **DHCPOFFER**
- **DHCPREQUEST**
- **DHCPACK**

Usage of Network Utilities





Identifying and Mitigating Security Risks

I gained knowledge on how to secure, troubleshoot, and monitor the performance of various components that comprise a Linux system.

Monitoring Linux Processes

I learned the default action of the “15 SIGTERM kill signal” is to kill a process that has been highlighted by the administrator.

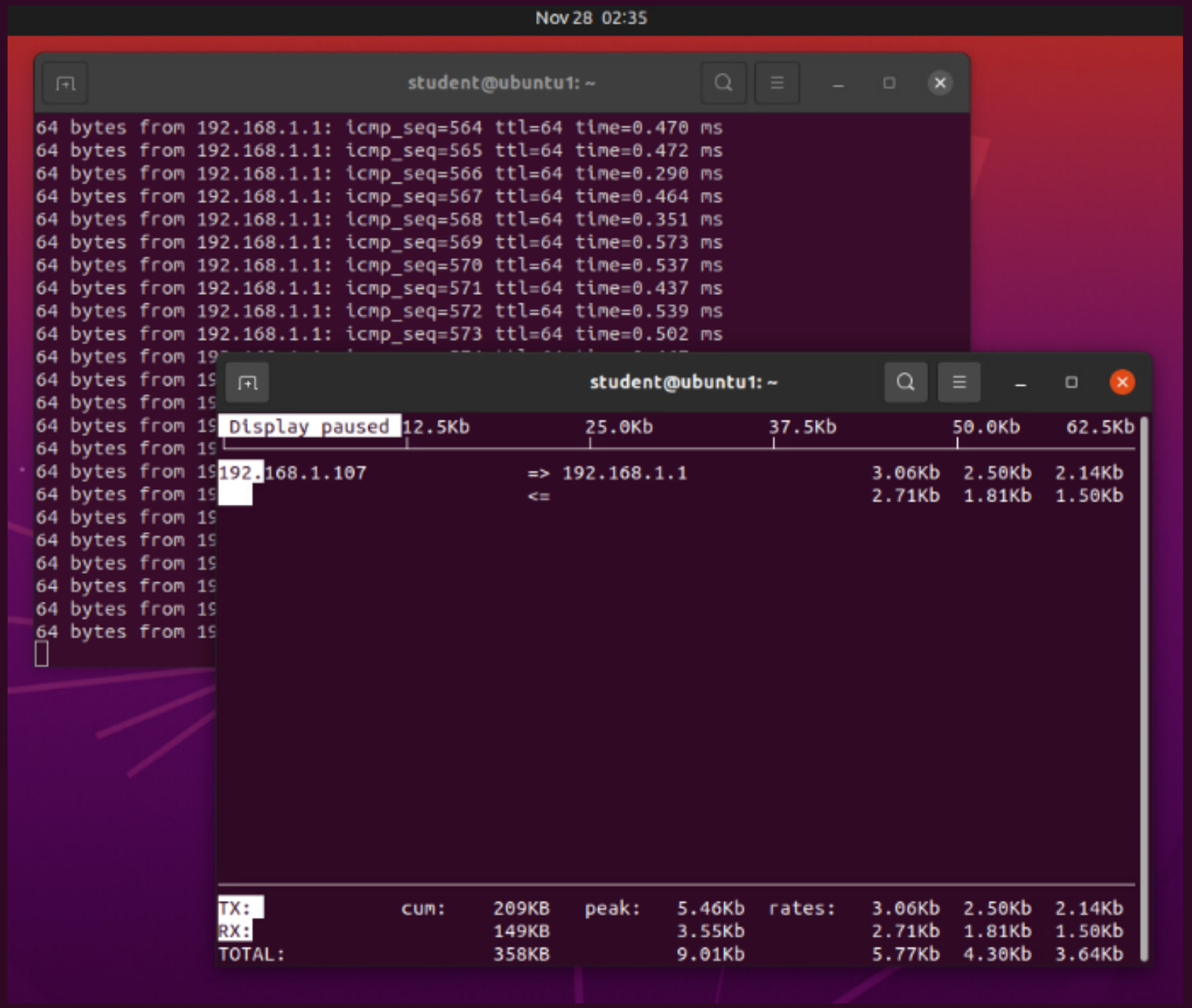
I also learned to utilize the System Monitor window to complete various tasks such as sorting processes by CPU usage in the format of a percentages. The process with the highest percentage of CPU usage on my machine was the gnome-shell (GUI) process.

Monitoring User Activities

I learned to monitor user activity in various scenarios. In one scenario I was able to conclude the flag value displayed in the output was “S”, which means “Super User”.

Typing sudo in front of a command permits a user to act as super user, which allows users to run commands with privileges that only the root user has been given.

Monitoring Network Bandwidth Usage



Challenges

- **One of the biggest challenges I faced was familiarizing myself with the amount of information required to manage an operating system.**
  Operating systems are filled with a lot of individual components and retaining knowledge of commands, individualized tasks, customizable options, and packages to manage was cumbersome when first starting my journey with the management of an operating system.
- **Another obstacle was getting accustomed to a Linux environment.**
  I am much more familiar with a windows environment and a change to the Linux operating system required a bit of a learning curve for myself.

Career Skills Obtained

The major career benefit of this course is the curriculum which is structured according to the CompTIA Linux+ certification guidelines.

I was able to obtain the following skills:

- **Working knowledge of the CLI and GUI for the Linux operating system**
- **Ability to create a virtual machine**
- **Run various administrative tasks within the Linux operating system such as monitoring various forms of activity, troubleshooting, managing groups/users, navigating the Linux File structure, creating shell scripts, configuring networks and many more.**

Conclusion

This project demonstrates that I can describe key features of the Linux operating system, create Linux shell scripts, navigate the Linux file system, run various administrative tasks, create basic network configurations, create a virtual machine with the Linux operating system for the IoT, and identify/mitigate security risks.